Hypothesis tests for more than two means and for correlation



Overview

Review/practice hypothesis test for more than two means

Hypothesis tests for correlation

If there is time: theories of hypothesis testing

Announcement

Homework 7 has been posted!

It is due on Gradescope on Sunday November 8th at 11pm

The midterm make-up exam being held right now in Kline Tower room 1105B!

- **If you had a Dean's Extension for the midterm exam please go to immediately to take the exam as this is the only time the make-up exam will be offered**

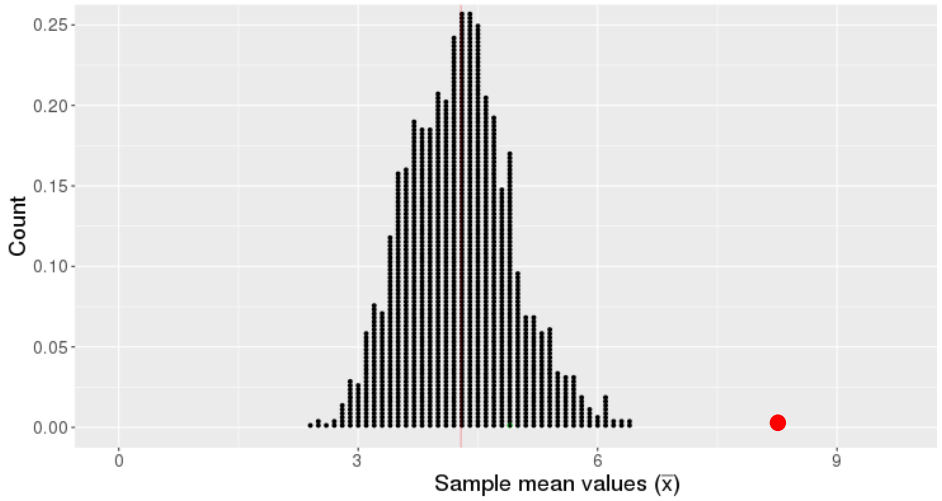
We aim have the midterm exams graded by Sunday

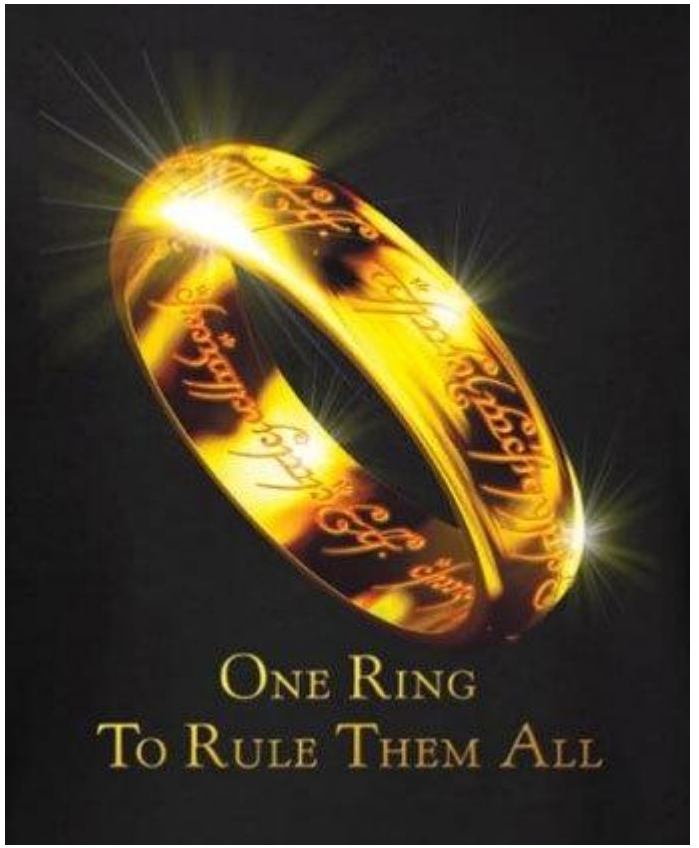
Review: hypothesis tests for more than two means

The logic of hypothesis tests…

There is only one hypothesis test!

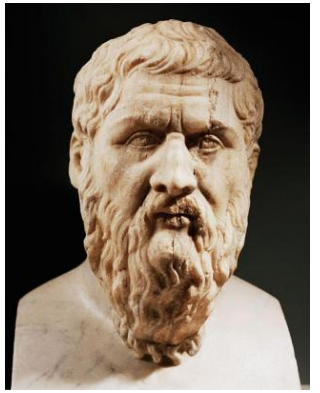
H_0: μ = 4.2





Just follow the 5 hypothesis tests steps!

Five steps of hypothesis testing

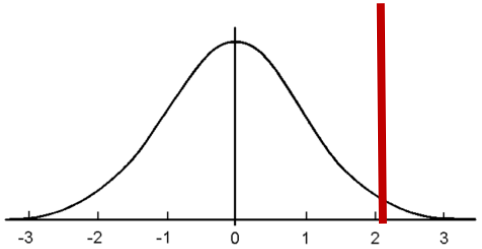


1. State H_0 and H_A
   - Assume Gorgias (H_0) was right



2. Calculate the actual observed statistic

3. Create a **null distribution** of statistics that are consistent with H_0
   - i.e., a distribution of statistics that we would expect if Gorgias is right

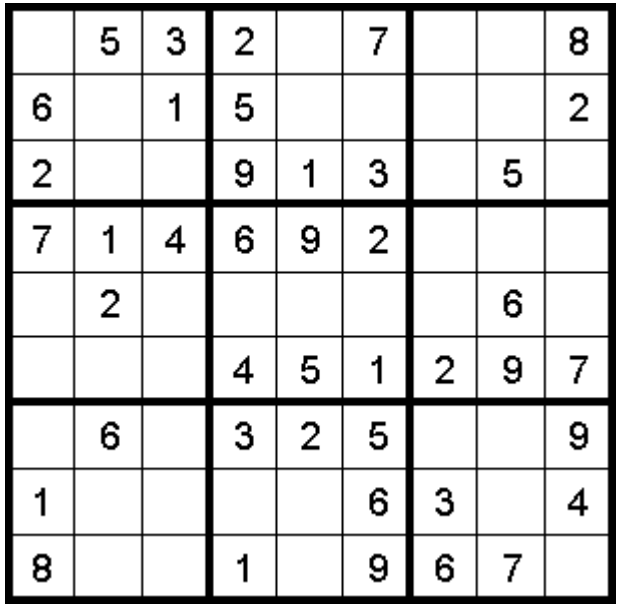


4. Get the probability we would get a statistic more than the observed statistic from the null distribution
   - p-value



5. Make a judgement
   - Assess whether the results are statistically significant

Comparing more than two means



	5	3	2		7			8
6		1	5					2
2			9	1	3		5	
7	1	4	6	9	2			
	2						6	
			4	5	1	2	9	7
	6		3	2	5			9
1					6	3		4
8			1		9	6	7	

Example from last class: comparing more than two means

A group of Hope College students wanted to see if there was an association between a student's major and the time it takes to complete a small Sudoku-like puzzle

	5	3	2		7			8
6		1	5					2
2			9	1	3		5	
7	1	4	6	9	2			
	2						6	
			4	5	1	2	9	7
	6		3	2	5			9
1					6	3		4
8			1		9	6	7	

They grouped majors into four categories:

- Applied science (as)
- Natural science (ns)
- Social science (ss)
- Arts/humanities (ah)

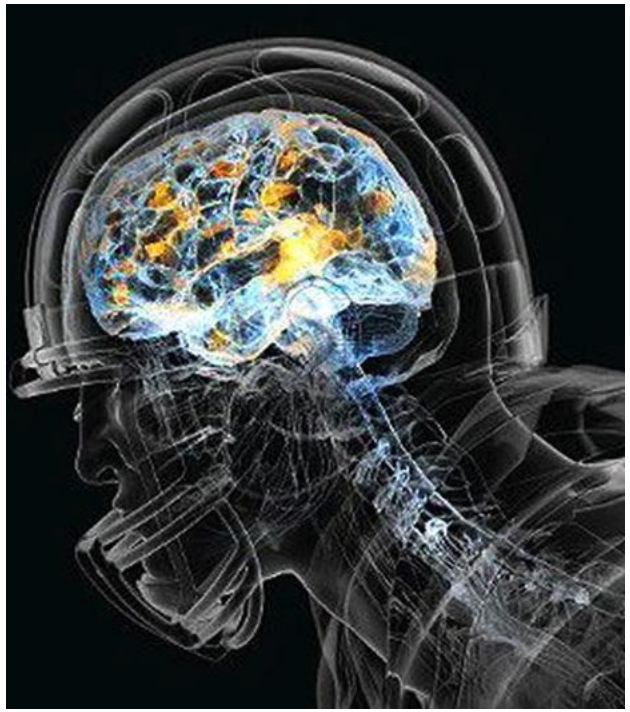


We tested:

- $\mathbf{H_0}$: $\mu_{as} = \mu_{ns} = \mu_{ss} = \mu_{ah}$
- $\mathbf{H_A}$: $\mu_i \neq \mu_j$ for at least one pair of fields of study

Does playing football affect brain size?

A study Singh et al (2014) published in the Journal of the American Medical Association (JAMA) examined the relationship between football and concussions on the brain.



The study included three groups with n = 25 participants in each group:

- Healthy controls who had never played football (HC)
- Football players with no history of concussions (FNC)
- Football players with a history of concussions (FC)

The researchers wanted to assess if there was a difference in the average hippocampus volume between these groups.

1. State the null and alternative hypotheses

$\mathbf{H_0}$: $\mu_{HC} = \mu_{FNC} = \mu_{FC}$

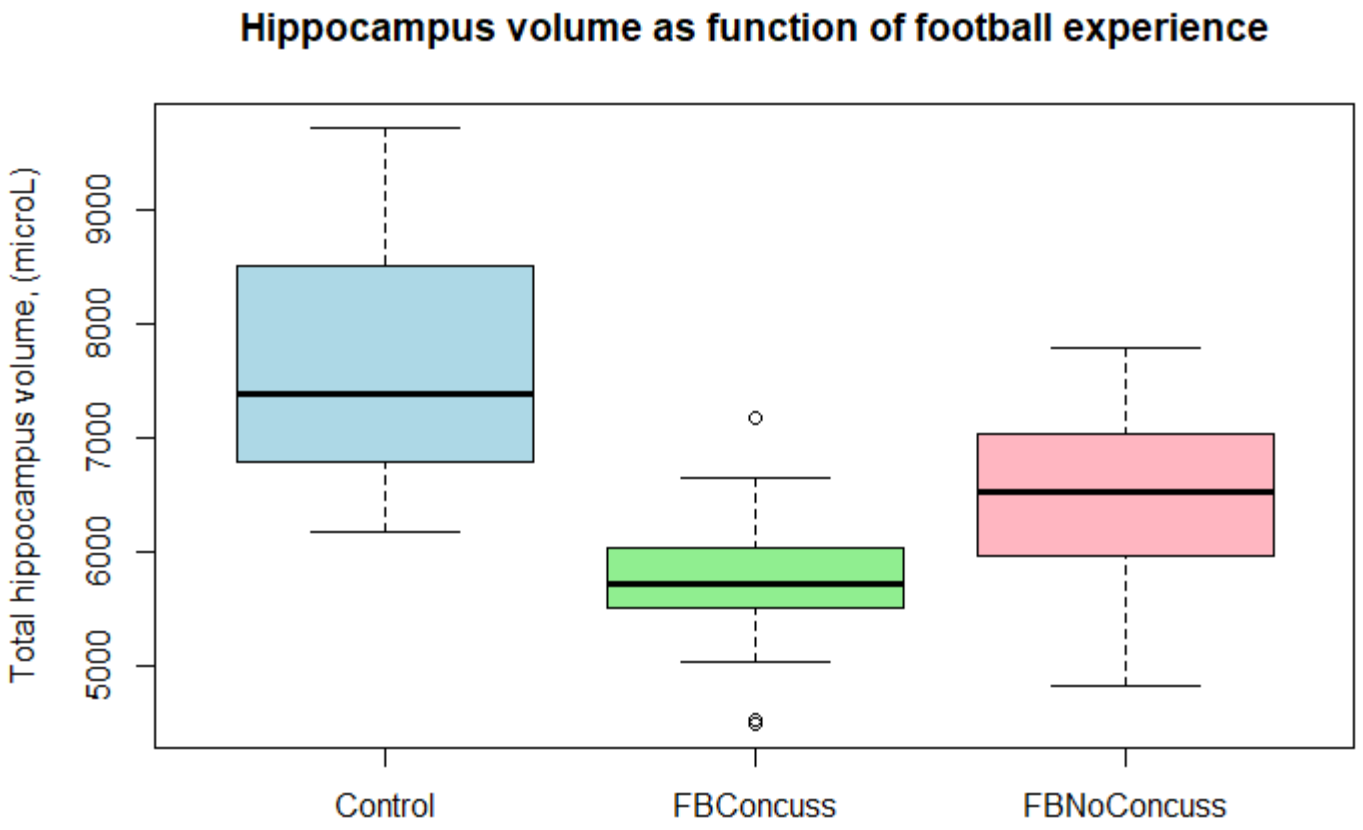
$\mathbf{H_A}$: $\mu_i \neq \mu_j$ for at least one pair of groups

What should we do next?

Let's plot the data first…

	Group	Hipp
1	Control	8510
2	FBNoConcuss	6405
3	Control	6220
4	Control	6780
5	FBNoConcuss	5155

Step 2a: Plot the data



boxplot(Hipp ~ Group)

What should we do next?

Comparing multiple means

There are many possible statistics we could use. A few choices are:

1. Group range statistic:

   $\max \bar{x} - \min \bar{x}$

2. Mean absolute difference (MAD):

   $(|\bar{x}_{as} - \bar{x}_{ns}| + |\bar{x}_{as} - \bar{x}_{ss}| + |\bar{x}_{as} - \bar{x}_{ah}| + |\bar{x}_{ns} - \bar{x}_{ss}| + |\bar{x}_{ns} - \bar{x}_{ah}| + |\bar{x}_{ss} - \bar{x}_{ah}|)/6$

3. F statistic:

$$F = \frac{\text{between-group variability}}{\text{within-group variability}}$$

Using the MAD statistic

Mean absolute difference (MAD):

$$(|\bar{x}_{as} - \bar{x}_{ns}| + |\bar{x}_{as} - \bar{x}_{ss}| + |\bar{x}_{as} - \bar{x}_{ah}| + |\bar{x}_{ns} - \bar{x}_{ss}| + |\bar{x}_{ns} - \bar{x}_{ah}| + |\bar{x}_{ss} - \bar{x}_{ah}|)/6$$

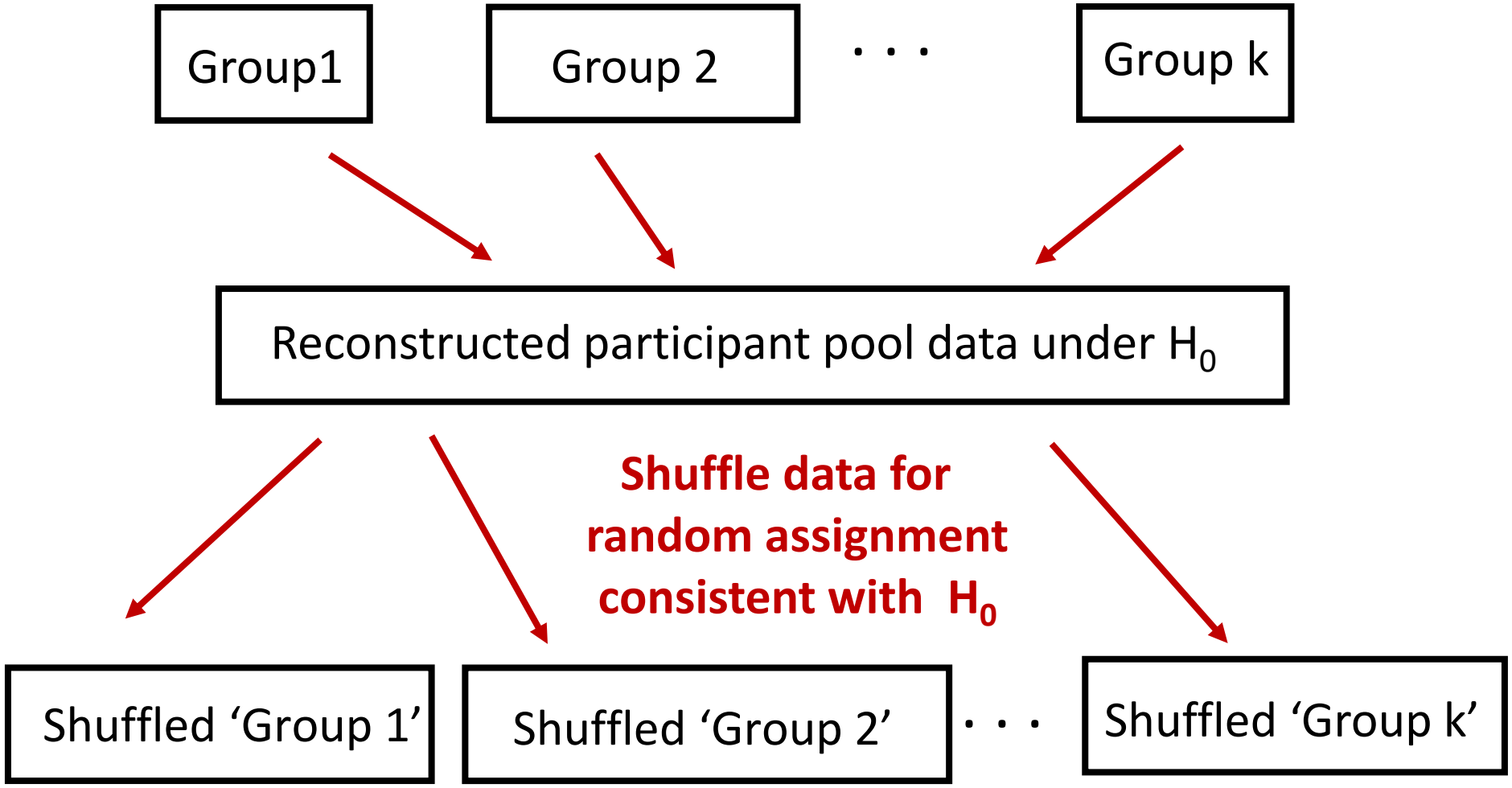
```
obs_stat <-  get_MAD_stat(Hipp, Group)
```

MAD statistic = 1245.333

What is step 3?

How can we create the null distribution?

3. Create the null distribution!



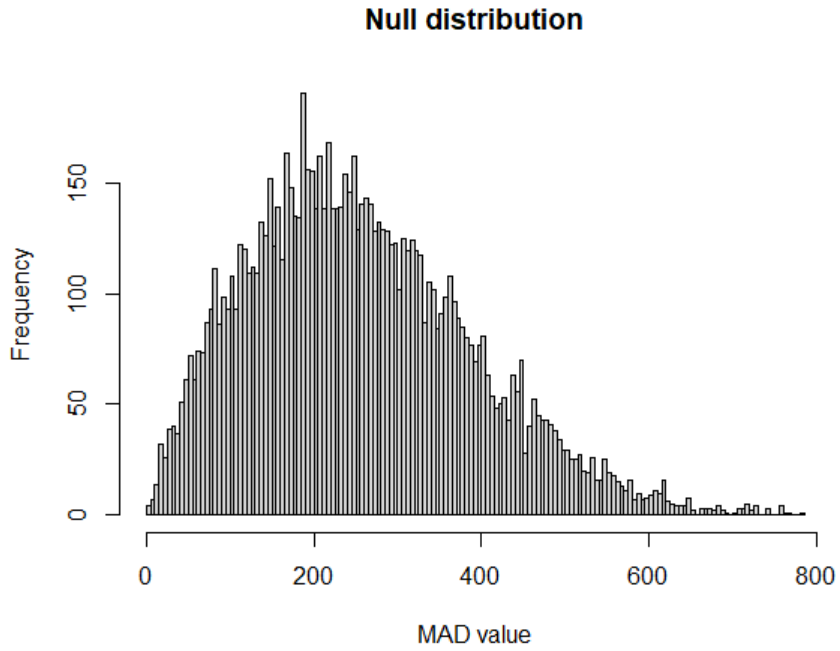
Compute MAD statistic from shuffled groups

Creating the null distribution

Q: How can we create a full null distribution?

```
null_dist <- do_it(10000) * {
      shuffled_concussions <- shuffle(Group)
      get_MAD_stat(Hipp, shuffled_concussions)
}

# visualize the null distribution
hist(null_dist, breaks = 200)
abline(v = obs_stat, col = "red")
```



Let’s try the rest in R…



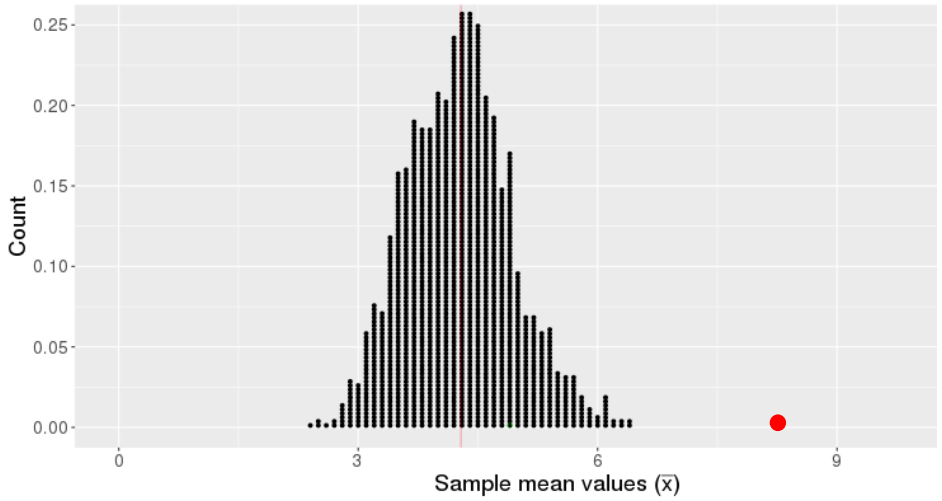


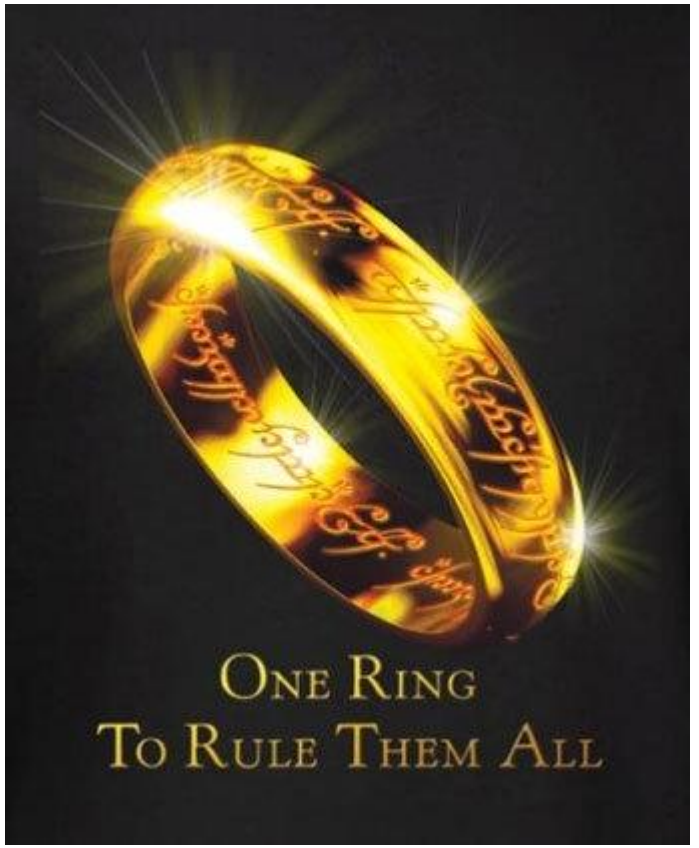
Hypothesis tests for correlation

The logic of hypothesis tests…

There is only one [hypothesis test](#)!

H_0: μ = 4.2





Just follow the 5 hypothesis tests steps!

Hypothesis tests for correlation

Is there a positive correlation between the amount of sugar in a cereal and the number calories?





What is the population parameter and the statistic of interest?

Significance tests for correlation

Let's look at data from 30 randomly selected cereals

	Calories	Sugars
AppleJacks	117	15.0
Boo Berry	118	14.0
Cap’n Crunch	144	16.0
Corn Flakes	101	3.0

What is the first step we should do for running a hypothesis test?

Hypothesis testing for correlation

1. Write down the null and alternative in symbols and words

2. Load the data and compute the observed statistic:

```
load("cereal.Rda")
```

3. Let’s extract the calories and carbohydrates from the data frame

```
calories <- cereal$Calories
sugar <- cereal$Sugars
```

Let's try it in R!

Step 2: What is the observed statistic?

- Also say whether you think you will be able to reject the null hypothesis based on a plot of your data

Step 3: Create the null distribution

- To start with: how we can create one point in the null distribution?
  - Hint: think about shuffling the data

Step 4: What is the p-value that you get?

Step 5: What decision would you make?



Homework preview

Homework 7 – Part 4: 1969 Vietnam Draft

Question: Was the 1969 Vietnam War draft "fair" in the sense that everyone, regardless of the date they were born, was equally likely to be drafted?



date	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
1	305	86	108	32	330	249	93	111	225	359	19	129
2	159	144	29	271	298	228	350	45	161	125	34	328
3	251	297	267	83	40	301	115	261	49	244	348	157
4	215	210	275	81	276	20	279	145	232	202	266	165
5	101	214	293	269	364	28	188	54	82	24	310	56
6	224	347	139	253	155	110	327	114	6	87	76	10
7	306	91	122	147	35	85	50	168	8	234	51	12
8	199	181	213	312	321	366	13	48	184	283	97	105
9	194	338	317	219	197	335	277	106	263	342	80	43
10	325	216	323	218	65	206	284	21	71	220	282	41
11	329	150	136	14	37	134	248	324	158	237	46	39
12	221	68	300	346	133	272	15	142	242	72	66	314
13	318	152	259	124	295	69	42	307	175	138	126	163
14	238	4	354	231	178	356	331	198	1	294	127	26
15	17	89	169	273	130	180	322	102	113	171	131	320
16	121	212	166	148	55	274	120	44	207	254	107	96
17	235	189	33	260	112	73	98	154	255	288	143	304
18	140	292	332	90	278	341	190	141	246	5	146	128
19	58	25	200	336	75	104	227	311	177	241	203	240
20	280	302	239	345	183	360	187	344	63	192	185	135
21	186	363	334	62	250	60	27	291	204	243	156	70
22	337	290	265	316	326	247	153	339	160	117	9	53
23	118	57	256	252	319	109	172	116	119	201	182	162
24	59	236	258	2	31	358	23	36	195	196	230	95
25	52	179	343	351	361	137	67	286	149	176	132	84
26	92	365	170	340	357	22	303	245	18	7	309	173
27	355	205	268	74	296	64	289	352	233	264	47	78
28	77	299	223	262	308	222	88	167	257	94	281	123
29	349	285	362	191	226	353	270	61	151	229	99	16
30	164		217	208	103	209	287	333	315	38	174	3
31	211		30		313		193	11		79		100

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1	305	86	108	32	330	249	93	111	225	359	19	129
2	159	144	29	271	298	228	350	45	161	125	34	328
3	251	297	267	83	40	301	115	261	49	244	348	157
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5	101	214	293	269	364	28	188	54	82	24	310	56
6	224	347	139	253	155	110	327	114	6	87	76	10
7	306	91	122	147	35	85	50	168	8	234	51	12
8	199	181	213	312	321	366	13	48	184	283	97	105
9	194	338	317	219	197	335	277	106	263	342	80	43
10	325	216	323	218	65	206	284	21	71	220	282	41
11	329	150	136	14	37	134	248	324	158	237	46	39
12									242	72	66	314
13									175	138	126	163
14	238	4	354	231	178	356	331	198	1	294	127	26
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16	121	212	166	148	55	274	120	44	207	254	107	96
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18	140	292	332	90	278	341	190	141	246	5	146	128
19	58	25	200	336	75	104	227	311	177	241	203	240
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24	59	236	258	2	31	358	23	36	195	196	230	95
25	52	179	343	351	361	137	67	286	149	176	132	84
26	92	365	170	340	357	22	303	245	18	7	309	173
27	355	205	268	74	296	64	289	352	233	264	47	78
28	77	299	223	262	308	222	88	167	257	94	281	123
29	349	285	362	191	226	353	270	61	151	229	99	16
30	164		217	208	103	209	287	333	315	38	174	3
31	211		30		313		193	11		79		100

The first date picked was Sept 14 (sequential number 258)

date	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
1	305	86	108	32	330	249	93	111	225	359	19	129
2	159	144	29	271	298	228	350	45	161	125	34	328
3	251	297	267	83	40	301	115	261	49	244	348	157
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28	77	299	223	262	308	222	88	167	257	94	281	123
29	349	285	362	191	226	353	270	61	151	229	99	16
30	164		217	208	103	209	287	333	315	38	174	3
31	211		30		313		193	11		79		100

The second date picked was April 24th (sequential number 115)

What is your Draft number?

date	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
1	305	86	108	32	330	249	93	111	225	359	19	129
2	159	144	29	271	298	228	350	45	161	125	34	328
3	251	297	267	83	40	301	115	261	49	244	348	157
4	215	210	275	81	276	20	279	145	232	202	266	165
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30	164		217	208	103	209	287	333	315	38	174	3
31	211		30		313		193	11		79		100

1969 Vietnam Draft sorted by sequential date

Date	Sequential date	Draft number
Jan 1	1	305
Jan 2	2	159
Jan 3	3	251
Jan 4	4	215
Jan 5	5	101
Jan 6	6	224
Jan 7	7	306
Jan 8	8	199
Jan 9	9	194

1969 Vietnam Draft

In a perfectly “fair”, random lottery, what should be the value of the correlation coefficient between **draft number** and **sequential date** of birthday?

Homework 7

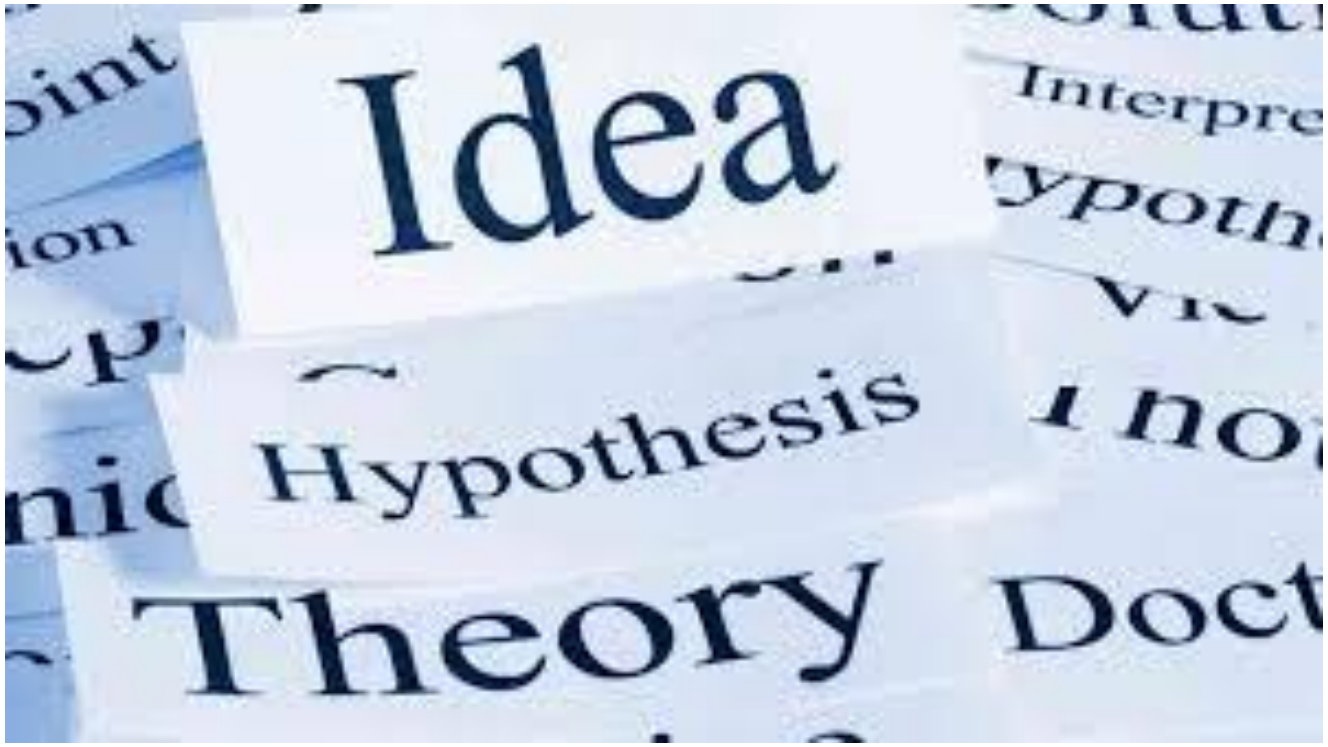
Use hypothesis testing to assess whether there is a correlation between sequential date and draft number

- i.e., was the draft really random?

Due 11pm on Sunday October 8h

If there is time…

Theories of hypothesis tests



Two theories of hypothesis testing

Null-hypothesis significance testing (NHST) is a hybrid of two theories:

1. Significance testing of Ronald Fisher
2. Hypothesis testing of Jezy Neyman and Egon Pearson



Fisher (1890-1962)



Neyman (1894-1981)

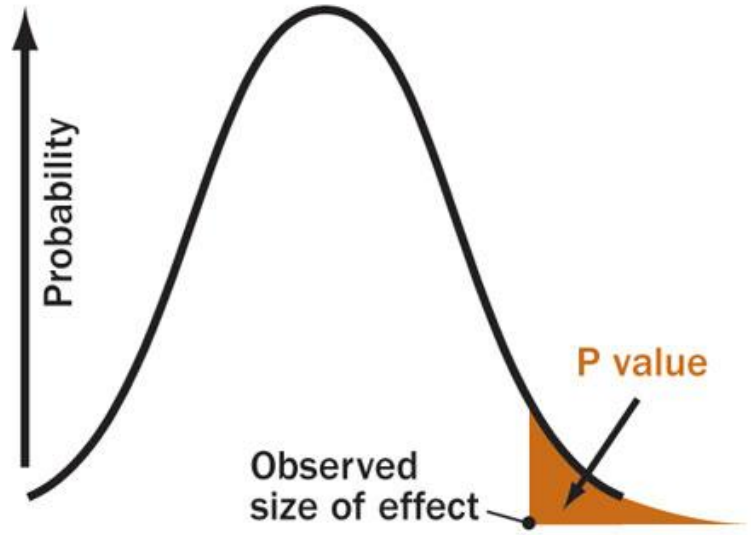


Pearson (1895-1980)

Ronald Fisher's significance testing

Views the p-value as strength of evidence against the null hypothesis

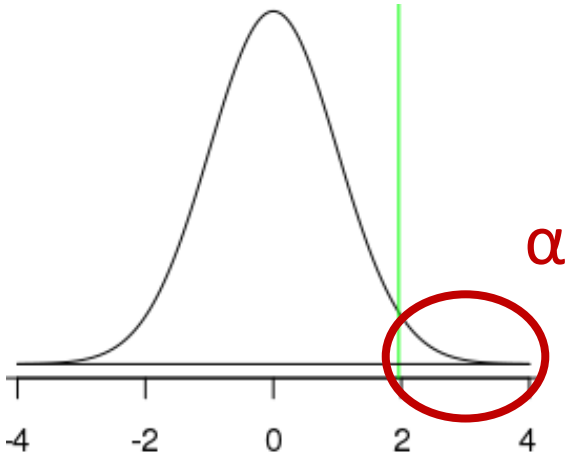
- P-values part of an on-going scientific process: tells the experimenter "what results to ignore"



Neyman-Pearson null hypothesis testing

Makes ***a formal decision*** in statistical tests

Null distribution



Reject H_0: if the observed sample statistic is beyond a fixed value

- i.e., reject H_0 if the p-value is less than some predetermined **significance level** α

Do not reject H_0: if the observed sample statistic is not beyond a fixed value. This means the test is inconclusive



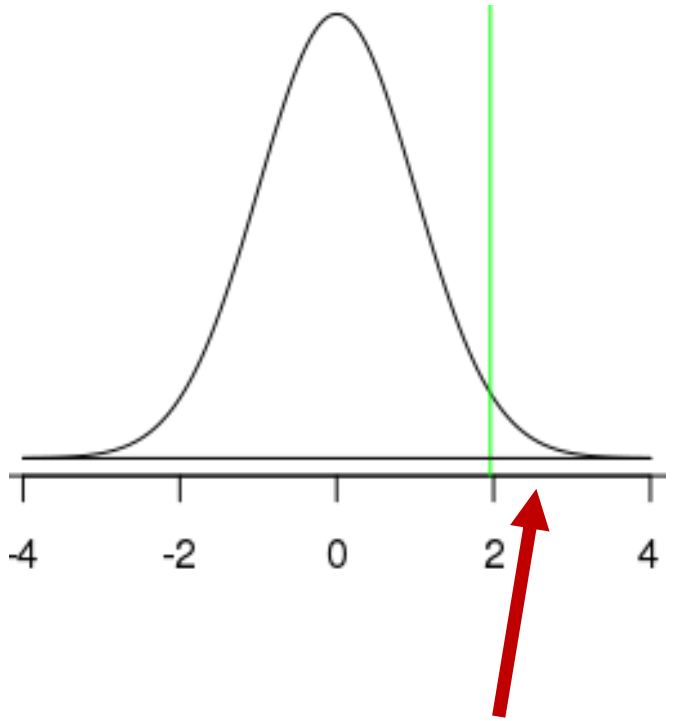
Neyman-Pearson frequentist logic

Type I error: incorrectly rejecting the null hypothesis when it is true

If Neyman-Pearson null hypothesis testing paradigm was followed, then only ~5% of the time when the null hypothesis was true, would we falsely reject it (for $\alpha = 0.05$)

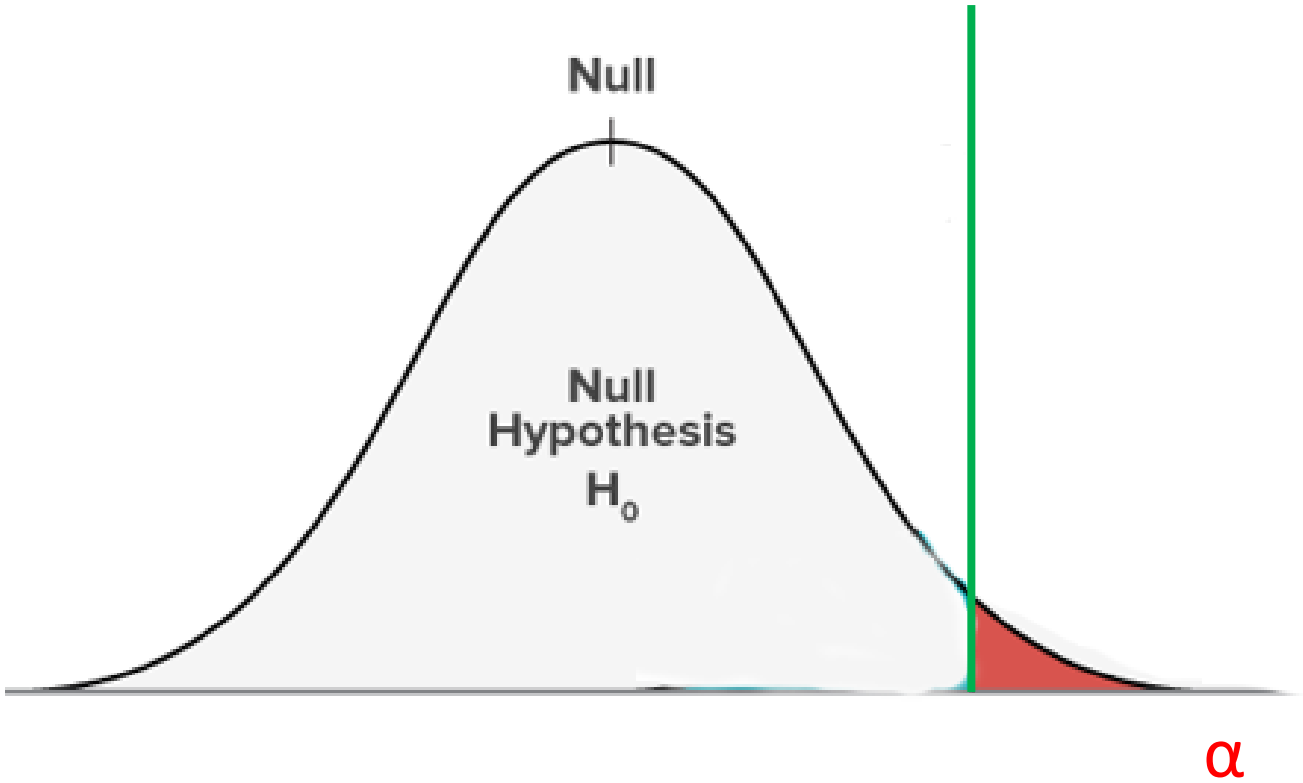
- i.e., we would only make type I errors 5% of the time

Null distribution

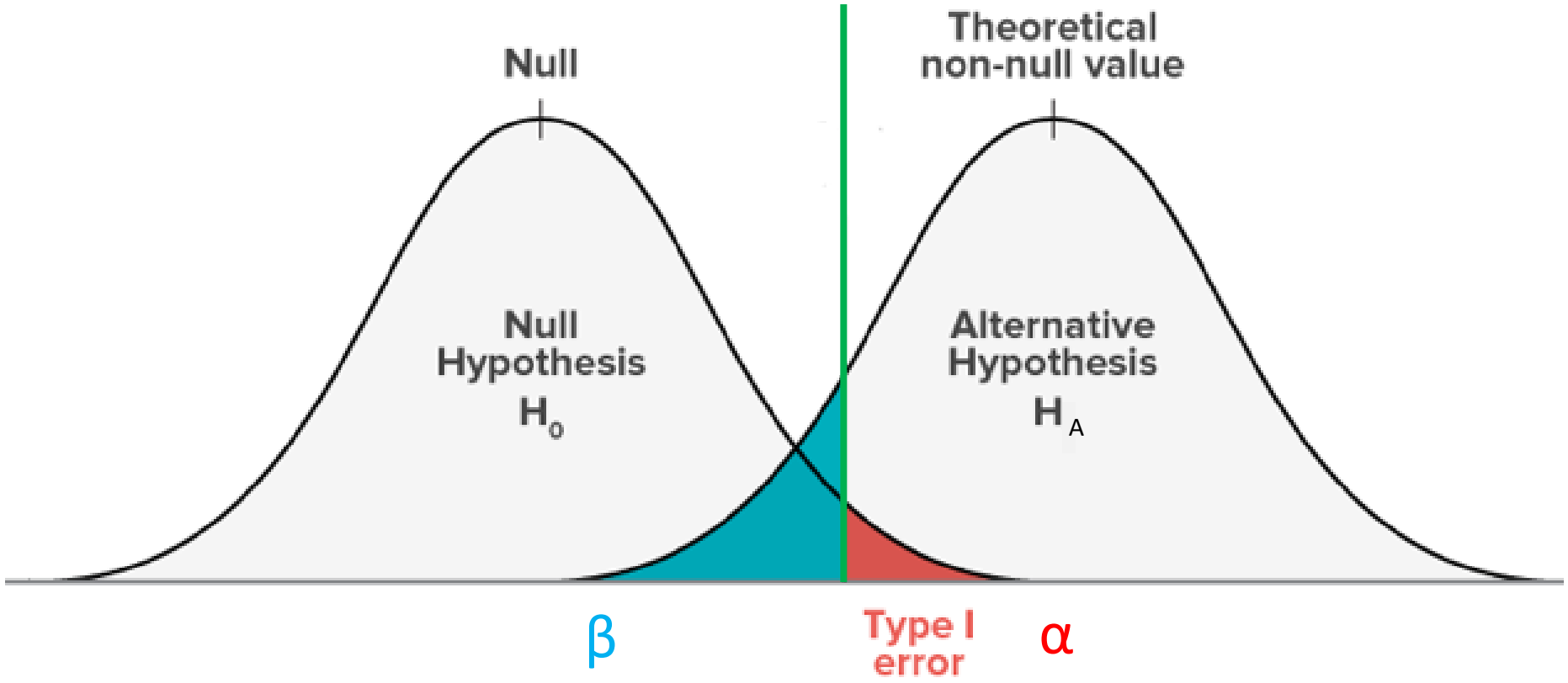


The null distribution is true but statistic landed here

Neyman-Pearson Frequentist logic



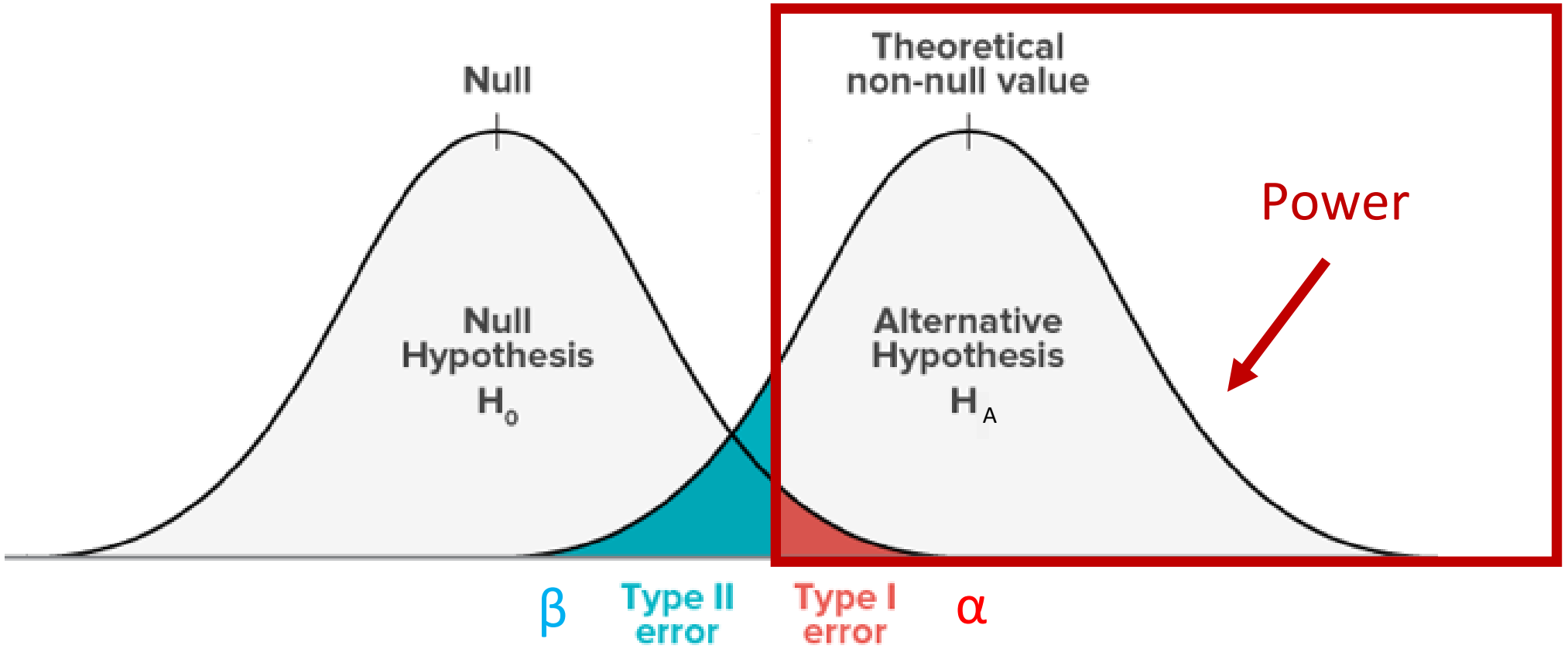
Neyman-Pearson Frequentist logic



Type II error: incorrectly failing to reject H_0 when it is false

- The rate at which we make type II errors is often denoted with the symbol β

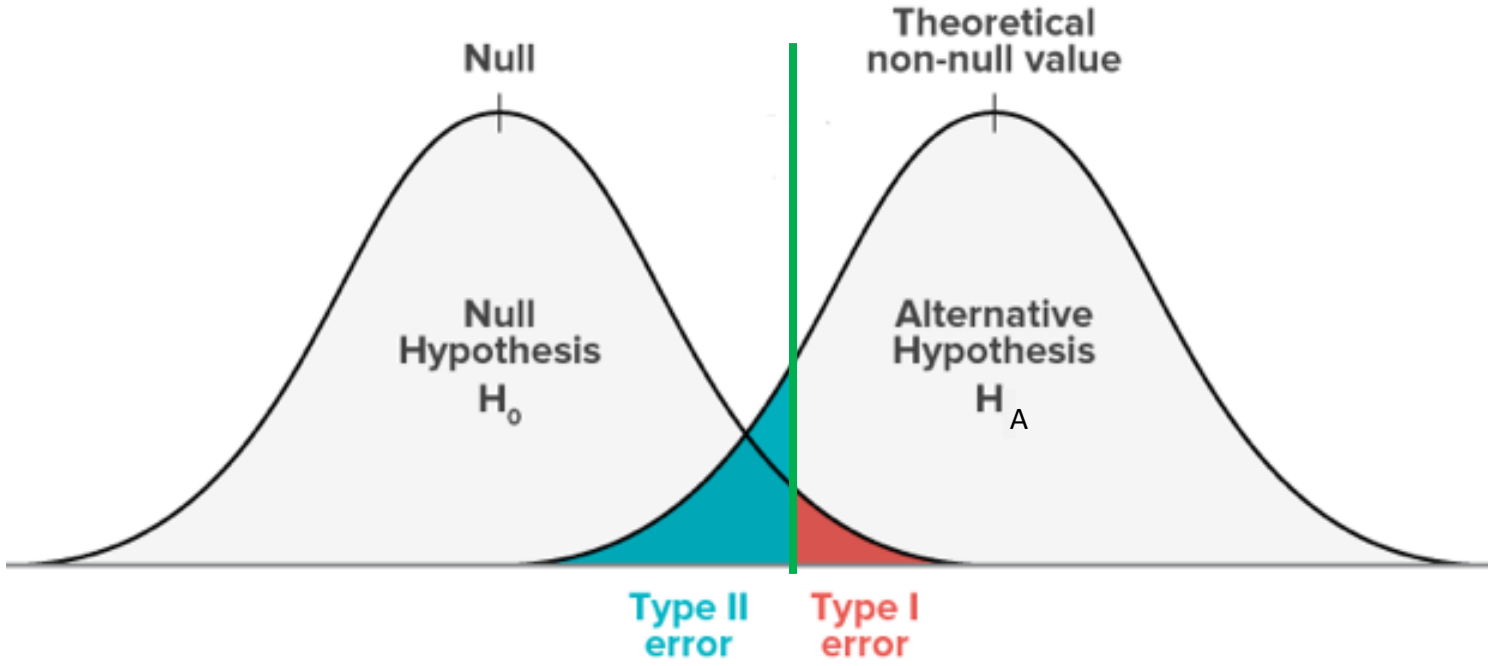
Neyman-Pearson Frequentist logic



The **power** of a test is the probability we reject the H_0 when it is **false**

- 1 - β
- For a fixed α level, it would be best to use the most powerful test

Type I and Type II Errors



	Reject H_0	**Do not reject H_0**
H_0 is true		
H_A is true (H_0 is false)		

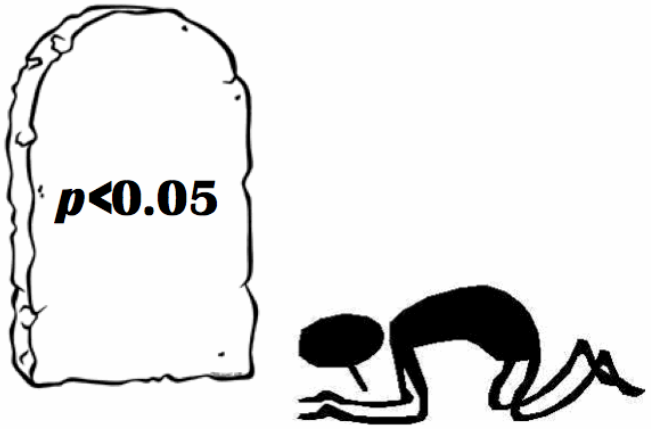
Problems with the NP hypothesis tests

Problem 1: we are interested in the results of a specific experiment, not whether we are right most of the time

- E.g., 95% of these statements are true:
  - Calcium is good for your heart, Paul is psychic, Buzz and Doris can communicate, …

Problem 2: Arbitrary thresholds for alpha levels

- P-value = 0.051, we don't reject H_0



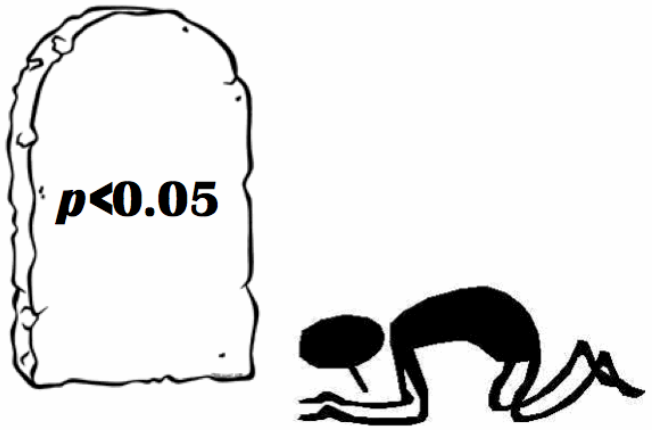
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Problem 2: Arbitrary thresholds for alpha levels

- P-value = 0.051, we don't reject H_0?



Problem 3: running many tests can give rise to a high number of type I errors

Genes and leukemia example

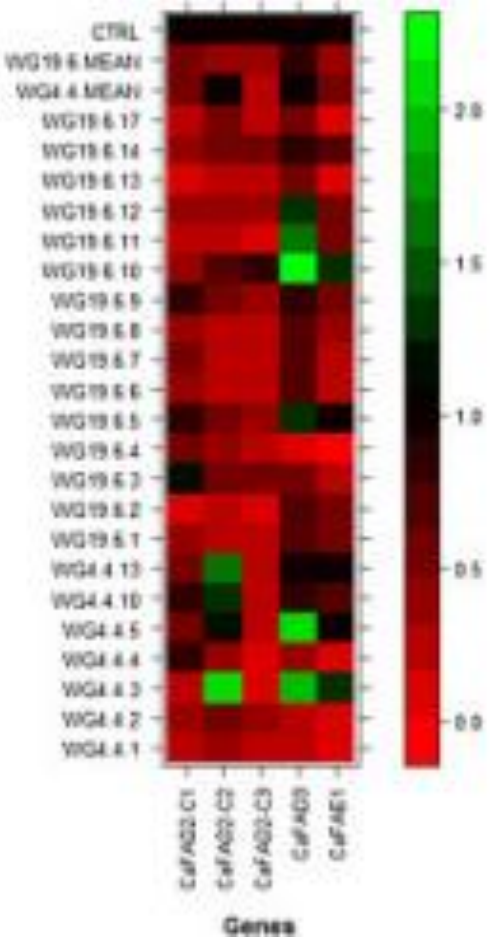
Scientists collected 7129 gene expression levels from 38 patients to find genetic differences between two types leukemia (L1 and L2)

Suppose there was no genetic differences between the types of leukemia

- H_0: $\mu_{L1} = \mu_{L2}$ is true for all genes

Q: If each gene was tested separately using a significance level of $\alpha = 0.05$, approximately how many type I errors would be expected?

- A: 7129 x 0.05 = 356



Side note: correcting for multiple comparisons

There are methods that try to correct for running multiple hypothesis tests

The ***Bonferroni correction*** is one way that controls the probability of ***any*** hypothesis test giving a type I error

- i.e., controls the familywise error rate (no type I errors for any of the tests run)

It works by dividing the initial α level by the number of tests run

- E.g., $\alpha = 0.05/7129 = 0.000007$
- All p-values need to be below this level to be considered statistically significant
- This can lead to many type II errors (Type II error: failure to reject H_0 when it is false)

The problem of multiple testing

For $\alpha = 0.05$, ~5% of all published research findings should incorrectly reject the null hypothesis

Publication bias (file drawer effect): Generally positive results are more likely to be published, so if you read the literature, the proportion of incorrect results could be greater than 5%

Essay

Why Most Published Research Findings Are False

John P. A. Ioannidis

The Earth Is Round ($p < .05$)

Jacob Cohen

After 4 decades of severe criticism, the ritual of null hypothesis significance testing—mechanical dichotomous decisions around a sacred .05 criterion—still persists. This article reviews the problems with this practice, including

sure how to test H_0, chi-square with Yates's (1951) correction or the Fisher exact test, and wonders whether he has enough power. Would you believe it? And would you believe that if he tried to publish this result without a

[American Statistical Association's Statement on p-values]

Some thoughts…

Better to have hypothesis tests than none at all. Just need to think carefully and use your judgment.



Report effect size in most cases – i.e., confidence intervals

Report the p-values rather than accept/reject H_0

- i.e., report $p = 0.023$ not $p < 0.05$

Replicate findings (perhaps in different contexts) to make sure you get the same results



Be a good/honest scientists and try to get at the Truth!